Healthcare Risk Prediction — Machine Learning vs Neural Networks

1. Problem Understanding

Diabetes is one of the most common health issues today, and many people are not aware of their condition until it becomes serious. Because of this, predicting diabetes at an early stage can help in taking preventive measures such as improving diet, increasing physical activity, or seeking medical advice.

In this assignment, the main goal is to build two different models to predict whether a person is likely to have diabetes or not:

- A Machine Learning model using Logistic Regression
- A Neural Network model using deep learning

2. Data Exploration Insights

Key Observations from the Dataset

- **Dataset Size:** The dataset contains around six thousand records with multiple features
- **Features Included:** patient_id, age, bmi, exercise_hours_per_week, smoker, blood_pressure, and diabetes
- **Data Quality:** The dataset is mostly clean with no missing values
- **Class Distribution:** The number of diabetic patients is significantly lower compared to non-diabetic patients, indicating a class imbalance problem

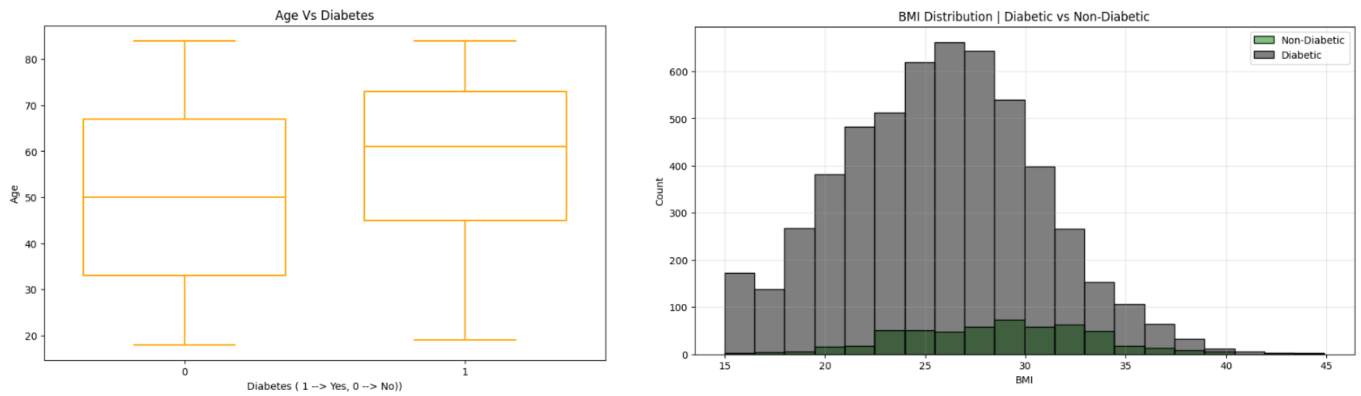
This imbalance can affect model performance, especially in predicting the minority class.

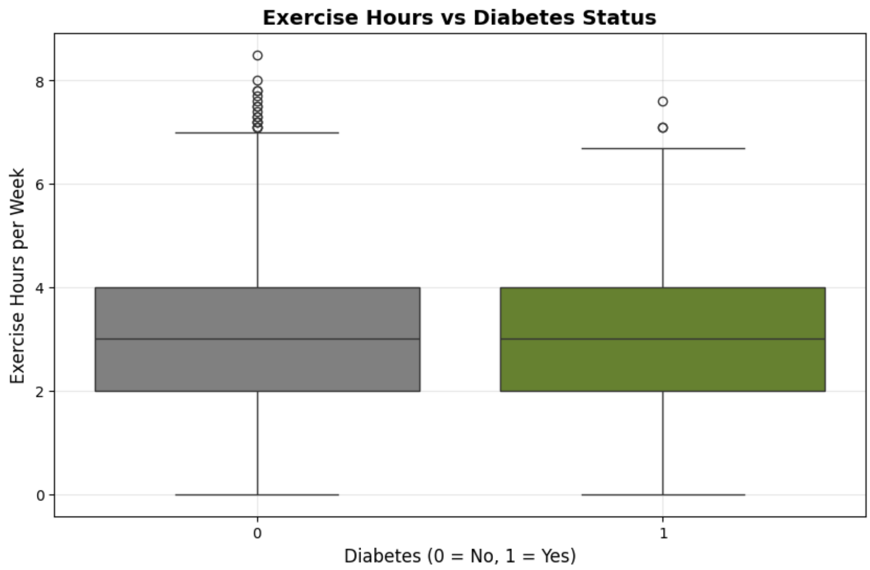
Important Features Influencing Prediction

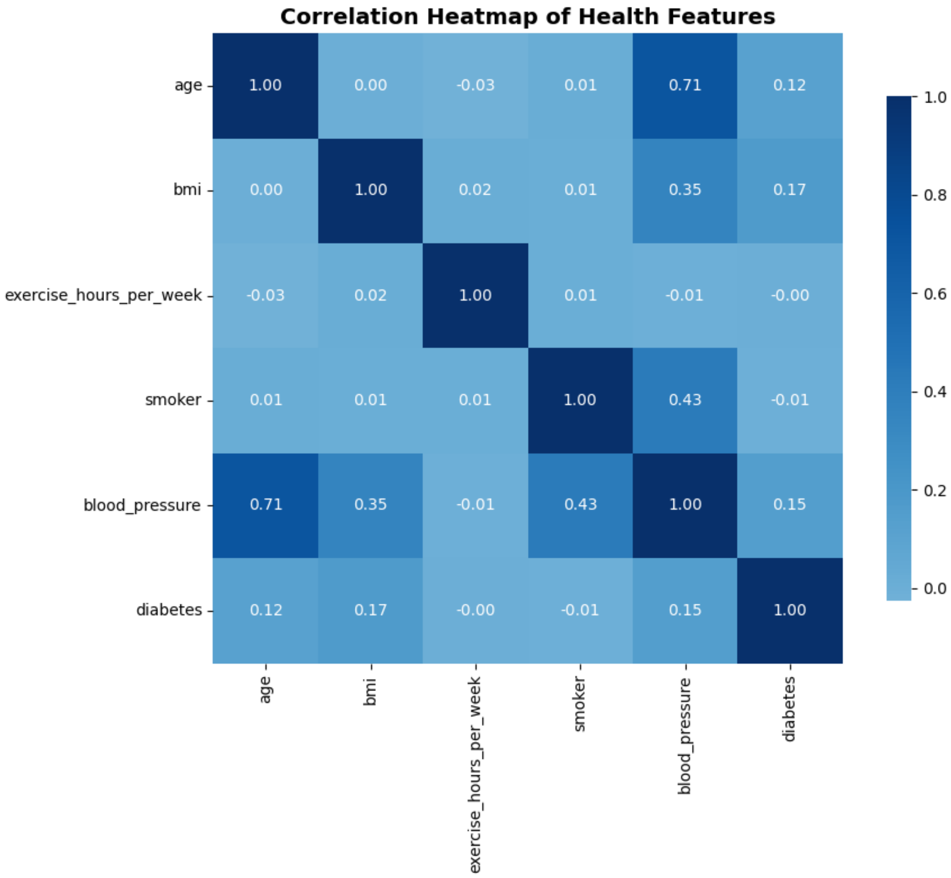
Based on initial analysis and visualizations:

- **Age:** Older individuals tend to have a higher chance of diabetes
- **BMI:** Higher BMI values are strongly associated with diabetic cases
- **Blood Pressure:** Elevated blood pressure shows a noticeable impact
- **Exercise:** Lower physical activity is linked to higher risk
- **Smoking:** Has some influence but is less significant compared to other features

These features play an important role in determining the final prediction.







Key insights from the Corelation Heatmap

- Blood Pressure increases with Age
- People with high Blood pressure have high BMI
- People doing more exercise per week have low diabetes risk
- Feature smoking shows weak correlation

3. Model Implementation

Machine Learning Model: Logistic Regression

The first model implemented was Logistic Regression, which is a simple and widely used classification algorithm.

- Preprocessing:
  Features and target variable were separated. The dataset was split into training and testing sets. Feature scaling was applied using StandardScaler.
- Handling Imbalance:
  Since the dataset is imbalanced, the parameter class_weight='balanced' was used. This helps the model give more importance to the minority class (diabetic patients).
- Training:
  The model was trained on the processed training data and predictions were made on the test set.

Neural Network Architecture

A basic Artificial Neural Network (ANN) was implemented using TensorFlow/Keras.

- Preprocessing:
  To handle class imbalance more effectively, SMOTE (Synthetic Minority Oversampling Technique) was applied to generate additional samples for the minority class.
- Model Layers:
  - Input layer with all features
  - Three hidden layers using ReLU activation
  - Output layer using Sigmoid activation for binary classification
- Compilation:
  The model was compiled using:
  - Adam optimizer
  - Binary cross-entropy loss function
- Training:
  The model was trained for 100 epochs, and performance was monitored using accuracy metrics.

4. Model Evaluation

Performance Metrics
Both models were evaluated using standard classification metrics:

- Accuracy
- Precision

- Recall
- F1 Score

Logistic Regression

	precision	recall	f1-score	support
0	0.95	0.65	0.77	814
1	0.16	0.65	0.26	86
accuracy			0.65	900
macro avg	0.56	0.65	0.52	900
weighted avg	0.87	0.65	0.72	900

- Provides stable and consistent results
- Performs well in identifying diabetic patients when balanced weights are used
- Slightly higher false positives due to class balancing

Neural Network Architecture

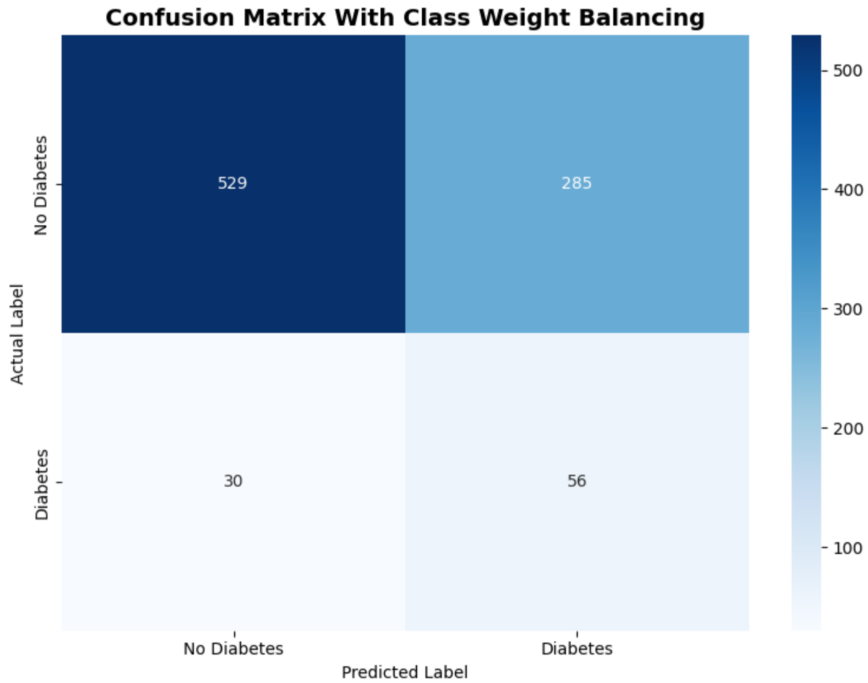
	precision	recall	f1-score	support
0	0.93	0.69	0.79	814
1	0.14	0.49	0.22	86
accuracy			0.67	900
macro avg	0.53	0.59	0.50	900
weighted avg	0.85	0.67	0.73	900

- Shows improved overall accuracy in some cases
- Performs better after applying SMOTE
- Capable of learning more complex relationships

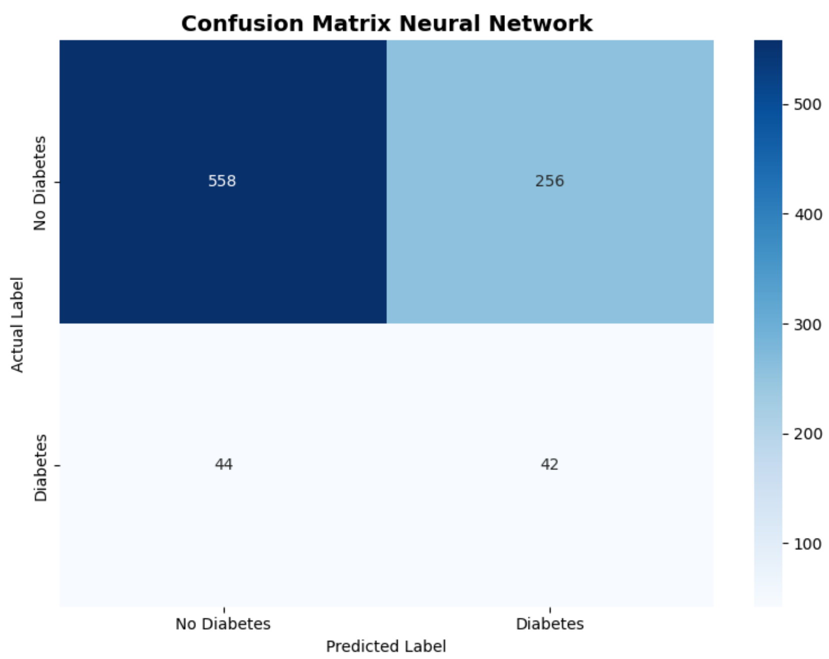
Visualizations

- **Confusion Matrix:**
  Helps in understanding correct vs incorrect predictions

Logistic Regression



Neural Network Architecture



5. Comparative Analysis

Logistic Regression may be preferred when:

- The goal is to **maximize recall** for diabetic patients
- Model interpretability is important
- Faster and simpler implementation is required
- Computational resources are limited

Neural Networks may be preferred when:

- The dataset has complex patterns and relationships
- Better overall performance is required
- There is enough data available for training
- Further tuning and optimization can be applied

Both the models used here have their own advantages and limitations. The Logistic Regression model worked well as a starting point and gave stable results. It is simple to implement and easy to understand, which makes it useful when we want to clearly know how the predictions are being made.

The Neural Network model, on the other hand, is more flexible and can capture more complex patterns in the data. With proper tuning, it has the potential to perform better than traditional models, although it requires more effort in terms of training and parameter adjustment.

Based on the results obtained, Logistic Regression was able to identify diabetic cases effectively, especially after handling the class imbalance. However, the Neural Network showed a better overall balance in terms of accuracy and prediction performance.

To conclude, if the focus is mainly on not missing diabetic patients, Logistic Regression can be a good and reliable choice. But if we are aiming for better overall performance and are willing to spend more time on tuning the model, then Neural Networks can be considered a better option.